

Motion-Conditioned Topological Artificial Intelligence for Fragmentation-Risk Forecasting in Unmanned Aerial Vehicle Networks

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Ercan Erkalkan: Conceptualization, Methodology, Software, Validation.
Formal analysis, Investigation, and Data curation.
Writing—original draft, Writing—review & editing, and Visualization.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have influenced the work reported in this article.

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Data and code availability

Code, configurations, simulated benchmark data, the derived real-flight trace, and generated artifacts are available from the following records:

Repository: <https://github.com/ErcanErkalkan/FANET-TopoGNN>

Software archive: <https://doi.org/10.5281/zenodo.20226053>

Source forestry field dataset (CC BY 4.0):

<https://doi.org/10.5281/zenodo.14701641>

MILUV three-UAV UWB dataset:

<https://doi.org/10.25452/figshare.plus.28386041.v1>

AERPAAW Dataset 22/23 records:

<https://aerpaw.org/dataset/>

WiNES UAV-to-UAV 60 GHz channel data:

<https://github.com/wineslab/uav-to-uav-60-ghz-channel-model>

An author-identifying-metadata-free copy is supplied separately for double-anonymous review.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the author used OpenAI GPT-based tooling for language refinement and local reproducibility cross-checks. The author reviewed and verified the resulting manuscript and code changes and takes responsibility for the submitted work.